INCH-POUND

MIL-PRF-19500/661
18 August 1998

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTORS, N-CHANNEL, SILICON
TYPES 2N7444, 2N7434, 2N7391 AND 2N7392
JANTXVR; AND JANSR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an N-channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event effects), power transistor intended for use in high density power switching applications. Two levels of product assurance are provided for each device type specified in MIL-PRF-19500, with avalanche energy maximum rating (E_{AS}) and maximum avalanche current (I_{AS}).

1.2 Physical dimensions. See figure 1, TO-254AA.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

Type	P_T 1/	P_T T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ 3/	I_{D2} T_C = +100°C	I_S 2/	I_{DM} 4/	T_{op} and T_{STG}	V_{ISO} 70,000 ft altitude
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A(pk)	°C	V dc
2N7444	250	3.0	200	200	±20	35.0	25.0	35.0	140	-55	NA
2N7434	250	3.0	250	250	±20	31.0	19.0	31.0	124	to	NA
2N7391	250	3.0	400	400	±20	22.0	14.0	22.0	88	+150	400
2N7392	250	3.0	500	500	±20	18.0	11.7	18.0	72		500

1/ Derate linearly 2.0 W/ °C for T_C > +25°C;

2/ $I_D = \sqrt{\frac{T_J \max - T_C}{(R_{\theta JC}) x (r_{DSon} \text{ at } T_J \max)}}$

3/ I_{D1} may be limited by pin diameter.

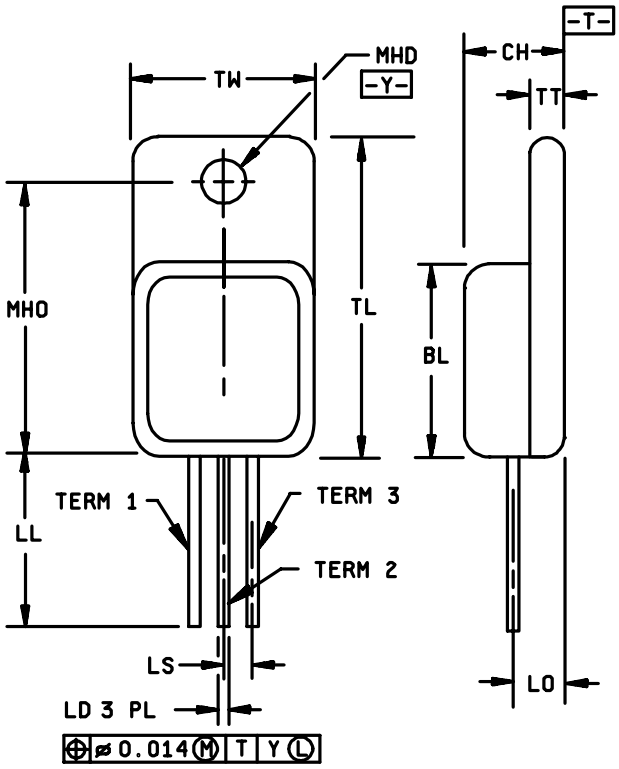
4/ I_{DM} = 4 X I_{D1}; as calculated by footnote 2/.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.535	.545	13.59	13.89
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.510	.570	12.95	14.48
LO	.150 BSC		3.81 BSC	
LS	.150 BSC		3.81 BSC	
MHD	.139	.149	1.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	20.32
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.89
Term 1	Drain			
Term 2	Source			
Term 3	Gate			



NOTES:
1. Dimensions are in millimeters.
2. Inches equivalents are given for general information only.
3. Refer to applicable symbol list.
4. Dimensioning and tolerating are in accordance with ANSI Y14.5M.
5. All terminals are isolated from case.

FIGURE 1. Physical dimensions for TO-254AA.

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1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ I_D = 1.0 mA dc		Max I_{DSS1} V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(ON)}$ 1/ V_{GS} = 12 V dc		$R_{\theta JC}$ max	E_{AS} at I_{D1}	I_{AS}
					T_J = +25°C at I_{D2}	T_J = +150°C at I_{D2}			
	V dc	V dc		µA dc	ohm	ohm	°C/W	mJ	A
		Min	Max						
2N7444	200	2.5	4.5	50	0.070	0.150	0.50	500	35.0
2N7434	250	2.5	4.5	50	0.110	0.250	0.50	500	31.0
2N7391	400	2.5	4.5	50	0.200	0.450	0.50	500	22.0
2N7392	500	2.5	4.5	50	0.320	0.700	0.50	500	18.0

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

I_{AS} - Rated avalanche current, nonrepetitive

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3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, MIL-HDBK-6100, and herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-PRF-19500 and herein. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.5).

3.3.2 Internal construction. Multiple chip construction is not be permitted to meet the requirements of this specification.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and table III herein. End-point electrical measurements shall be in accordance with applicable tests of table I, group A, subgroup 2 herein.

4.2.1.1 Single Event Effects. Design capability shall be tested on the initial qualification and thereafter whenever a major die design or process change is introduced. Figure 3 illustrates the design safe operation area. End-point electrical measurements shall be in accordance with table III.

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4.3 Screening (JANTXV and JANS levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTXV levels
1/	Method 3470, E_{AS} (see 4.5.4)	Method 3470, E_{AS} (see 4.5.4)
1/	Method 3161, thermal impedance (see 4.5.3)	Method 3161, thermal impedance (see 4.5.3)
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
9 2/	Subgroup 2 of table I herein. I_{GSS}, I_{DSS1}	Not applicable
10	MIL-STD-750, method 1042, test condition B	MIL-STD-750, method 1042, test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein ΔI_{GSSF1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 10 µA dc or ± 100 percent of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein
12	MIL-STD-750, method 1042, test condition A	MIL-STD-750, method 1042, test condition A
13	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 10 µA dc or ± 100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value $\Delta V_{GS(th)1}$ = ± 20 percent of initial value	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 10 µA dc or ± 100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value $\Delta V_{GS(th)1}$ = ± 20 percent of initial value

1/ Shall be performed anytime before screen 10.
2/ Shall be performed after E_{AS} test, thermal impedance test, and gate stress test.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Alternate flow is allowed for quality conformance inspection in accordance with of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein. End-point electrical measurements shall be in accordance with table I, group A, subgroup 2 herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500, and herein. End-point electrical measurements shall be in accordance with table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1051	Test condition G, 100 cycles.
B3	2075	See 3.3.2.
B3	2077	SEM qualification may be performed anytime prior to lot formation.
B4	1042	Test condition D. The heating cycle shall be 30 seconds minimum.
B5	1042	Accelerated steady-state reverse bias, condition B, V_{DS} = rated, T_A = +175°C, t = 24 hours minimum; or, T_A = +150°C, t = 48 hours minimum.
B5	1042	Accelerated steady-state reverse bias, condition A, V_{GS} = rated, T_A = +175°C, t = 120 hours minimum; or, T_A = +150°C, t = 240 hours minimum.
B5	2037	Bond strength (Al-Au die interconnects only); test condition A.
B6	3161	See 4.5.2.

4.4.2.2 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition G, 25 cycles.
B3	1042	Test condition D. The heating cycle shall be 30 seconds minimum.
B3	2037	Test condition A. All internal bond wires for each device shall be pulled separately.
B5 and B6	----	Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable of table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition A; weight = 10 pounds; t = 15 s.
C5	1001	Test condition C. V_{DS} = Rated; $I_{(ISO)}$ < 0.25 mA. (Not required for 2N7444 or 2N7434)
C6	1042	Test condition D. The heating cycle shall be 30 seconds minimum.

4.4.4 Group D Inspection. Group D inspection shall be conducted in accordance with table VII of MIL-PRF-19500 and table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in of MIL-STD-750.

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4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC(max)}$ = 0.50°C/W. The following parameter measurements shall apply:

a. Measuring current (I_M) ..... 10 mA.

b. Drain heating current (I_H) ..... 8 A minimum.

c. Heating time (t_H) ..... Steady-state (see MIL-STD-750, method 3161for definition).

d. Drain-source heating voltage (V_H) ..... 25 V.

e. Measurement time delay (t_{MD}) ..... 30 to 60 μs.

f. Sample window time (t_{SW}) ..... 10 μs maximum.

4.5.3 Thermal impedance ($Z_{\theta JC}$ measurements). The $Z_{\theta JC}$ measurements shall be performed in accordance with MIL-STD-750, method 3161. The maximum limit (not to exceed figure 2, thermal impedance curves and the group A, subgroup 2 limits) for $Z_{\theta JC}$ in screening (table VI of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure may be used in lieu of an in line monitor.

a. Measuring current (I_M) ..... 10 mA.

b. Drain heating current (I_H) ..... 8 A minimum.

c. Heating time (t_H) ..... 100 ms .

d. Drain-source heating voltage (V_H) ..... 25 V.

e. Measurement time delay (t_{MD}) ..... 30 to 60 μs.

f. Sample window time (t_{SW}) ..... 10 μs maximum.

4.5.4 Single pulse avalanche energy E_{AS}.

a. Peak current (I_{AS}) ..... $I_{AS(max)}$.

b. Peak gate voltage (V_{GS}) ..... 12 V.

c. Gate to source resistor (R_{GS}) ..... $25\Omega \leq R_{GS} \leq 200\Omega$.

d. Initial case temperature (T_C) ..... +25°C, +10°C, -5°C.

e. Inductance (L) ..... $\dfrac{\left[\dfrac{2E_{AS}}{(I_{D1})^2}\right]\left[(V_{BR}-V_{DD})\right]}{V_{BR}}$ *mH minimum.*

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Supply voltage (V_{DD}) ..... 50 V .

4.5.5 Gate stress test.

a. V_{GS} = 30 V minimum.

b. t = 250 μs minimum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.5.3	$Z_{\theta JC}$		0.39	°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V; I_D = 1 mA dc Bias condition C	$V_{(BR)DSS}$			
2N7444				200		V dc
2N7434				250		V dc
2N7391				400		V dc
2N7392				500		V dc
Gate to source voltage threshold	3403	$V_{DS} \geq V_{GS}$ I_D = 1 mA dc	$V_{GS(TH)1}$	2.5	4.5	V dc
Gate current	3411	V_{GS} = +20 and -20 V dc V_{DS} = 80 percent of rated V_{DS}	I_{GSS1}		± 100	nA dc
Drain current	3413	V_{GS} = 0 V dc, Bias condition C V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		50	µA dc
Static drain to source “ON”-state resistance	3421	V_{GS} = 12 V dc, condition A pulsed (see 4.5.1), I_D = I_{D2}	$r_{DS(on)1}$			
2N7444					0.070	ohm
2N7434					0.110	ohm
2N7391					0.200	ohm
2N7392					0.320	ohm
Static drain to source “ON”-state resistance	3421	V_{GS} = 12 V dc, condition A pulsed (see 4.5.1), I_D = I_{D1}	$r_{DS(on)2}$			
2N7444					0.075	ohm
2N7434					0.123	ohm
2N7391					0.210	ohm
2N7392					0.360	ohm
Forward voltage	4011	Pulsed (see 4.5.1), , I_D = I_{D1} V_{GS} = 0 V dc	V_{SD}			
2N7444					1.4	V dc
2N7434					1.4	V dc
2N7391					1.4	V dc
2N7392					1.8	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate current	3411	V_{GS} = +20 and -20 V dc Bias condition C, V_{DS} = 0	I_{GSS2}		± 200	nA dc
Drain current	3413	V_{GS} = 0 V; bias condition C V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		0.25	mA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = 12 V dc pulsed (see 4.51), $I_D = I_{D2}$	$r_{DS(on)3}$			
2N7444					0.140	ohm
2N7434					0.220	ohm
2N7391					0.390	ohm
2N7392					0.590	ohm
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)2}$	1.5		V dc
Low temperature operation:		$T_C = T_J = +55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)3}$		5.5	V dc
Subgroup 4						
Forward transconductance	3475	I_D = rated I_{D2}, V_{DD} = 15 V (see 4.5.1)	g_{FS}			
2N7444				12		S
2N7434				10		S
2N7391				6		S
2N7392				6		S
Switching time test	3472	$I_D = I_{D1}$, V_{GS} = 12 V dc R_G = 2.35Ω, V_{DD} = 50 percent of rated V_{DS}				
Turn-on delay time			$t_{d(on)}$			
2N7444					35	ns
2N7434					30	ns
2N7391					28	ns
2N7392					29	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Rise time			t_r			
2N7444					200	ns
2N7434					130	ns
2N7391					97	ns
2N7392					93	ns
Turn-off delay time			$t_{d(off)}$			
2N7444					150	ns
2N7434					100	ns
2N7391					120	ns
2N7392					90	ns
Fall time			t_f			
2N7444					150	ns
2N7434					90	ns
2N7391					72	ns
2N7392					59	ns
Subgroup 5	3474					
Safe operating area test (high voltage)		See figure 3 t_p = 10 ms minimum V_{DS} = 80 percent of maximum rated V_{DS}, ($V_{DS} \leq 200$)				
Electrical measurements		See table I, group A, subgroup 2				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			
2N7444					260	nC
2N7434					210	nC
2N7391					185	nC
2N7392					180	nC

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7 - Continued						
Gate to source charge			Q_{gs}			
2N7444					80	nC
2N7434					50	nC
2N7391					35	nC
2N7392					30	nC
Gate to drain charge			Q_{gd}			
2N7444					150	nC
2N7434					110	nC
2N7391					100	nC
2N7392					95	nC
Reverse recovery time	3473	di/dt ≤ 100 A/μs, V_{DD} ≤ 50 V, $I_D = I_{D1}$	t_{rr}			
2N7444					650	ns
2N7434					700	ns
2N7391					720	ns
2N7392					800	ns

1/ For sampling plan, see MIL-PRF-19500.

2/ This test is required for the following end-point measurements only (not intended for screen 13):

JANS - group B, subgroups 3 and 4
JANTX and JANTXV - group B, subgroups 2 and 3;

group C, subgroup 6;
group E, subgroup 1

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TABLE II. Group D inspection.

Inspection 1/, 3/, 4/	MIL-STD-750		Symbol	Preirradiation limits		Postirradiation limits		Unit
	Method	Conditions		R		R		
				Min	Max	Min	Max	
Subgroup 1								
Not applicable								
Subgroup 2		T_C = +25°C						
Steady-state total dose irradiation (V_{GS} bias) 2/	1019	V_{GS} = 12V V_{DS} = 0						
Steady-state total dose irradiation (V_{DS} bias) 2/	1019	V_{GS} = 0 V_{DS} = 80% of rated V_{DS} (pre-irradiation)						
End-point electricals								
Breakdown voltage, drain to source	3407	V_{GS} = 0 I_D = 1 mA Bias Cond. C	V_{BRDSS}					
2N7444				200		200		V dc
2N7434				250		250		V dc
2N7391				400		400		V dc
2N7392				500		500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$	V_{GSth1}					
2N7444				2.5	4.5	2.0	4.5	V dc
2N7434				2.5	4.5	2.0	4.5	V dc
2N7391				2.5	4.5	2.0	4.5	V dc
2N7392				2.5	4.5	2.0	4.5	V dc
Gate current	3411	V_{GS} = 20 V V_{DS} = 0 Bias Cond. C	I_{GSSF1}		100		100	nA dc
Gate current	3411	V_{GS} = -20 V V_{DS} = 0 Bias Cond. C	I_{GSSR1}		-100		-100	nA dc
Drain current	3413	V_{GS} = 0 Bias Cond. C V_{DS} = 80% of rated V_{DS} (pre-irradiation)	I_{DSS1}		50		50	μA dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/3/4/	MIL-STD-750		Symbol	Preirradiation limits		Postirradiation limits		Unit
	Method	Conditions		R		R		
				Min	Max	Min	Max	
Static drain to source on-state voltage	3405	V_{GS} = 12 V Cond. A pulsed (see 4.5.1) $I_D = I_{D2}$	V_{DSon1}					
2N7444					1.75		1.75	V dc
2N7434					2.09		2.09	V dc
2N7391					2.80		2.80	V dc
2N7392					3.744		3.744	Vdc
Forward voltage source to drain diode	4011	V_{GS} = 0 $I_D = I_{D1}$	V_{SD}					
2N7444					1.4		1.4	V dc
2N7434					1.4		1.4	V dc
2N7391					1.4		1.4	V dc
2N7392					1.8		1.8	V dc

1/ For sampling plan, see MIL-PRF-19500.
2/ Separate samples shall be pulled for each bias.
3/ Group D qualification may be performed anytime prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other detail specification utilizing the same die design.
4/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in it's qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.

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TABLE III. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			12 devices c = 0
Thermal shock (temperature cycling)	1051	Test condition G.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, group A, subgroup 2.	
Subgroup 2 1/			12 devices c = 0
Steady-state gate bias	1042	Condition B, 1,000 hours.	
Electrical measurements		See table I, group A, subgroup 2.	
Steady-state reverse bias	1042	Condition A, 1,000 hours.	
Electrical measurements		See table I, group A, subgroup 2.	
Subgroup 3			
Not applicable			
Subgroup 4			12 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 0.50 °C/W maximum. See 4.5.2.	
Subgroup 5			12 devices c = 0
Barometric Pressure	1001	V_{DS} = rated; $I_{(ISO)}$ < 0.25 mA Not required for 2N7444 and 2N7434	
Subgroup 6			3 devices c = 0
Electrical measurements 2/		IGSS1 and IDSS1 per table I, group A, subgroup 2.	
Single Event Effects (SEE) 3/ 4/ 5/	1080		
SEE irradiation		Fluence = 3E5 ± 20% ions/cm^2 Flux = 2E3 to 2E4 ions/cm^2/sec Temperature = 25 ± 5 °C	
2N7444		LET = 28 MeV-cm^2/mg range = 43 microns Energy = 285 MeV Insitu bias conditions: VDS = 200 V & VGS = -20 V LET = 37 MeV-cm^2/mg range = 39 microns Energy = 305 MeV Insitu bias conditions: VDS = 200 V & VGS = -10 V VDS = 180 V & VGS = -15 V VDS = 140 V & VGS = -20 V	

See footnotes at end of table.

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TABLE III. Group E inspection (all quality levels) for qualification only. - continued

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 6 - continued			3 devices c = 0
2N7434		LET = 28 MeV-cm^2/mg range = 43 microns Energy = 285 MeV Insitu bias conditions: VDS = 250 V & VGS = -20 V LET = 37 MeV-cm^2/mg range = 39 microns Energy = 305 MeV Insitu bias conditions: VDS = 250 V & VGS = -10 V VDS = 225 V & VGS = -15 V VDS = 210 V & VGS = -20 V	
2N7391		LET = 28 MeV-cm^2/mg range = 43 microns Energy = 285 MeV Insitu bias conditions: VDS = 325 V & VGS = -15 V LET = 37 MeV-cm^2/mg range = 39 microns Energy = 305 MeV Insitu bias conditions: VDS = 325 V & VGS = -10 V VDS = 275 V & VGS = -15 V	
2N7392		LET = 28 MeV-cm^2/mg range = 43 microns Energy = 285 MeV Insitu bias conditions: VDS = 375 V & VGS = -20 V LET = 37 MeV-cm^2/mg range = 39 microns Energy = 305 MeV Insitu bias conditions: VDS = 350 V & VGS = -10 V VDS = 325 V & VGS = -15 V VDS = 300 V & VGS = -20 V	
Electrical measurements 2/		IGSS1 and IDSS1 per table I, group A, subgroup 2.	

1/ A separate sample may be pulled for each test condition.

2/ Examine IGSS1 and IDSS1 before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions per table I, group A, subgroup 2, may be performed at the manufacturer's option.

3/ Group E qualification of single event effect testing may be performed prior to lot formation. Qualification may be extended to other performance specifications utilizing the same structurally identical die design.

4/ Device qualification to a higher level LET is sufficient to qualify all lower level LET's.

5/ The sampling plan applies to each bias condition.

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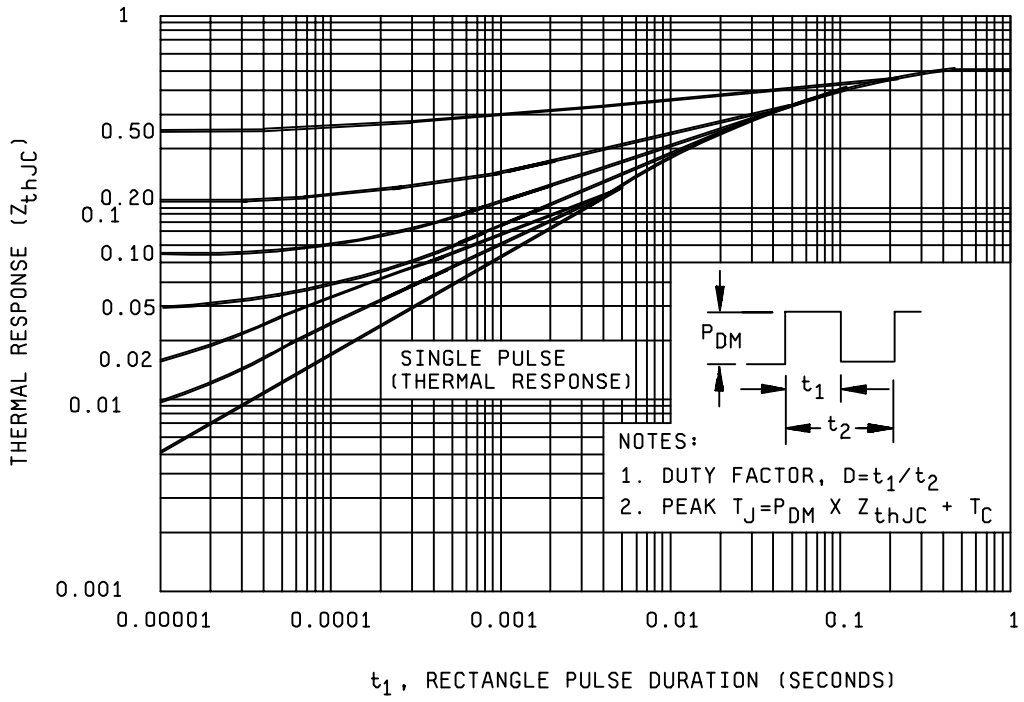


FIGURE 2. Thermal response curve.

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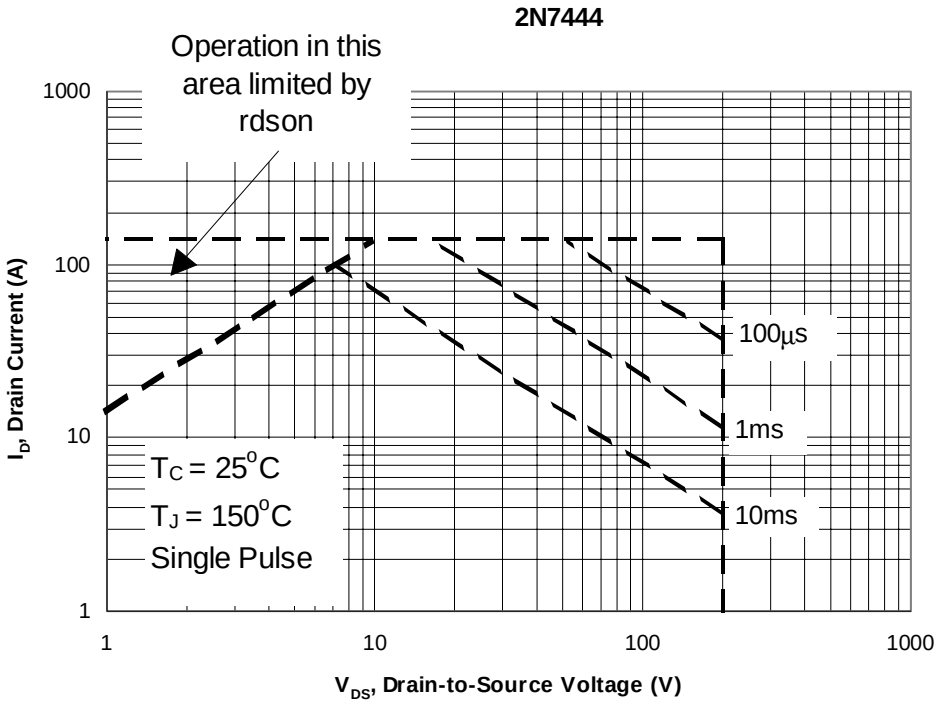


FIGURE 3. Safe operating area graph.

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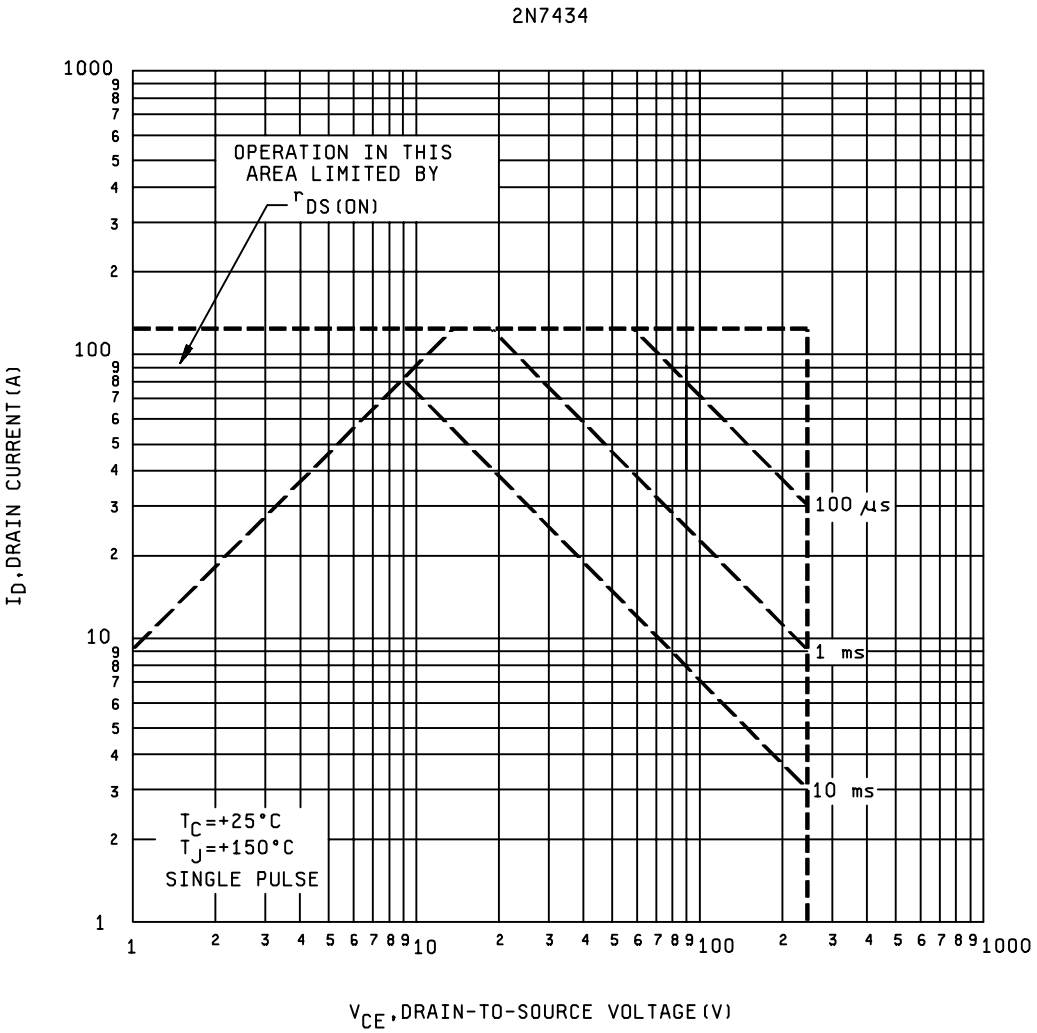


FIGURE 3. Safe operating area graph - continued

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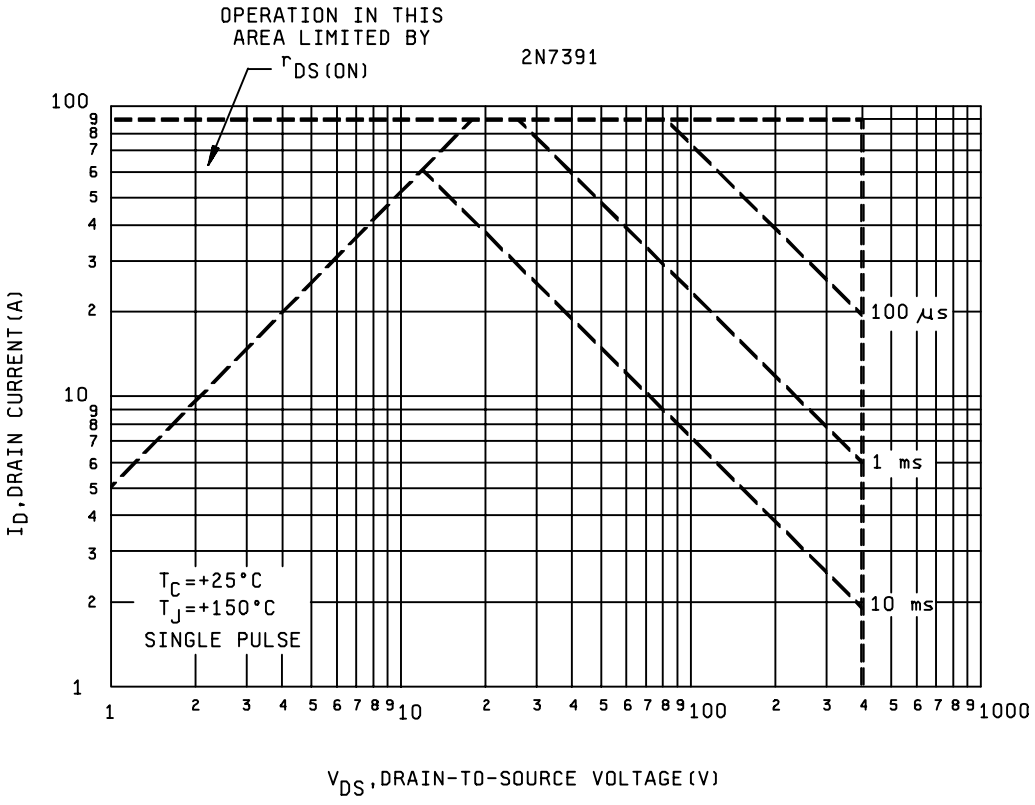


FIGURE 3. Safe operating area graph - continued.

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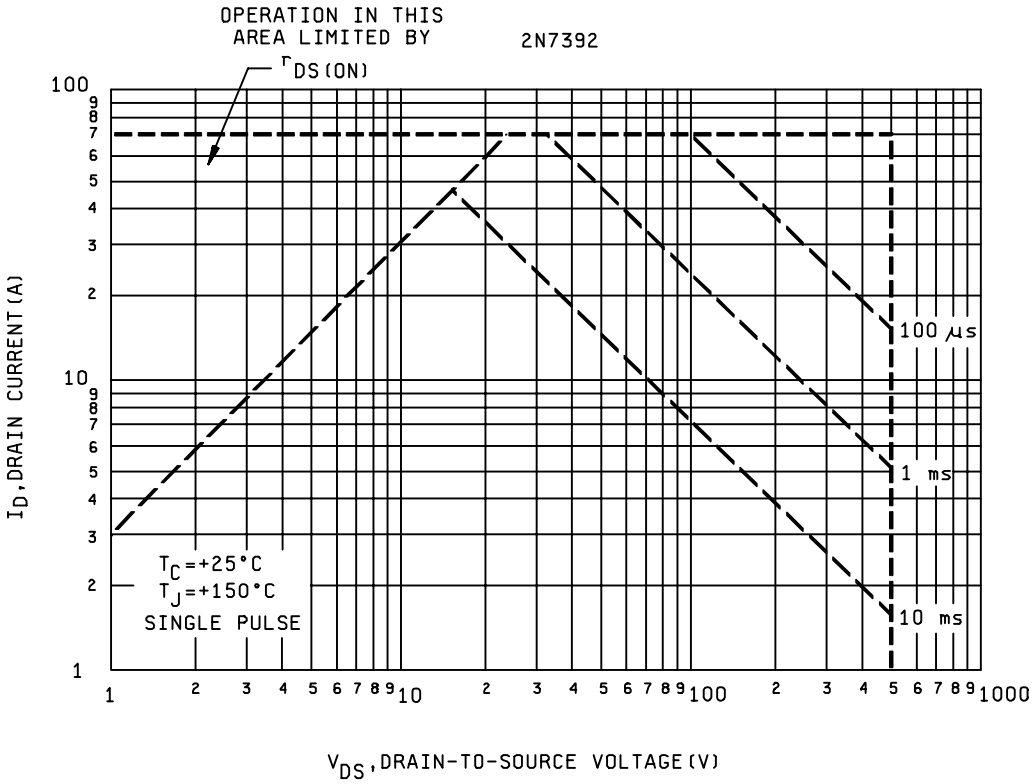


FIGURE 3. Safe operating area graph - continued

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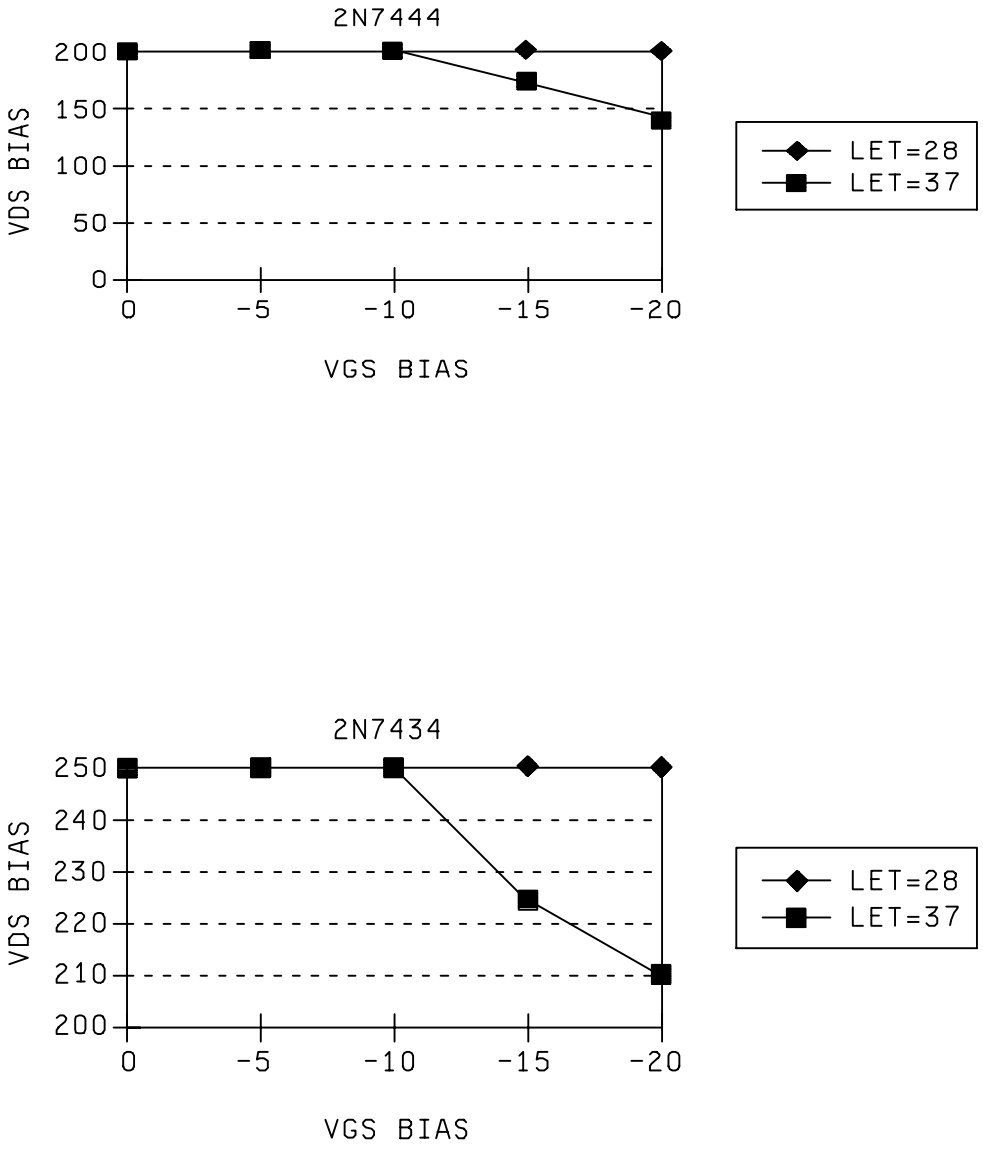
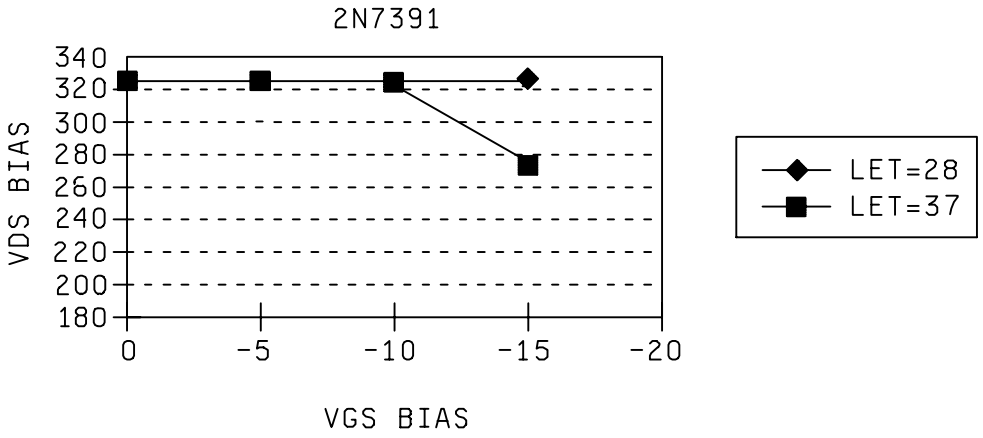


FIGURE 4. Single event effects safe operation area .

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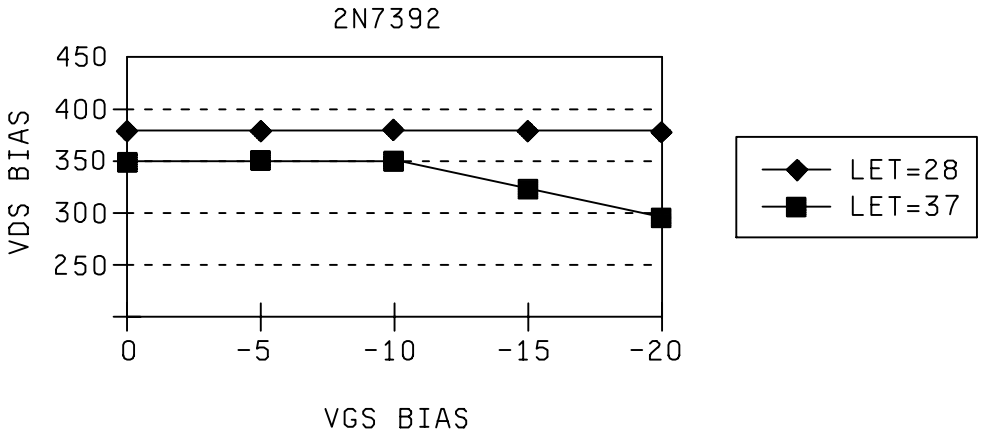


FIGURE 4. Single event effects safe operation area - continued.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129 and MIL-PRF-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable for the military PIN.

Preferred types Military PIN	Commercial PIN	
	TO254AA	
2N7444	IRHM7260SE	
2N7434	IRHM7264SE	
2N7391	IRHM7360SE	
2N7392	IRHM7460SE	

6.5 Ordering data. Acquisition documents may specify the material and finish (see 3.3.1).

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Preparing activity:
DLA - CC

(Project 5961-2005)

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/661	**2. DOCUMENT DATE** 18 August 1998

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED (TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTORS, N-CHANNEL, SILICON TYPES 2N7444, 2N7434, 2N7391 AND 2N7392 JANTXVR; AND JANSR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial DSN FAX EMAIL 614-692-0510 850-0510 614-692-6939 alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete

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